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4.2.3. City that Sold the Most Products

01:45

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Group the data by City and calculate the total quantity of products sold for each city.
- Find the city that sold the most products (based on the total quantity sold).

Sample Data :

Date,Product,Quantity,Price,City

2025-01-01,Product A,5,20,New York

2025-01-01,Product B,3,15,Los Angeles

2025-01-02,Product A,7,20,New York

2025-01-02,Product C,4,30,Chicago

2025-01-03,Product B,2,15,Chicago

2025-01-03,Product A,8,20,Los Angeles

2025-01-04,Product C,6,30,New York

2025-01-04,Product B,5,15,Los Angeles

2025-01-05,Product A,3,20,Chicago

2025-01-05,Product C,10,30,Los Angeles

Note:

The data cannot be displayed in the file. You can refer to the sample data provided for insights.

```
import pandas as pd

# Prompt the user for the file name
file_name = input()

# Load the data
df = pd.read_csv(file_name)

# Group by City and calculate total quantity
city_sales = df.groupby('City')['Quantity'].sum()

# Find the city with the highest total quantity sold
best_city = city_sales.idxmax()

# Display the result
print(f'City sold the most products: {best_city}')
```

The screenshot displays a code execution environment with the following components:

- Performance Metrics:**
 - Average time: 0.018 s (18.33 ms)
 - Maximum time: 0.036 s (36.00 ms)
- Test Results:**
 - 1 out of 1 shown test case(s) passed
 - 2 out of 2 hidden test case(s) passed
- Test Case Details:**
 - Test case 1:** 36 ms, with a 'Debug' button.
 - Expected output:** sales_data.csv, City sold the most products: Los Angeles
 - Actual output:** sales_data.csv, City sold the most products: Los Angeles
- Navigation:** Buttons for 'Terminal' and 'Test cases' at the bottom.